APPLICATIONS OF THE TRIANGULATION THEOREM FOR O-MINIMAL STRUCTURES

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Abstract.
In this work we discuss some applications of the triangulation theorem for O-minimal structures (proved in the previous seminar). We will have some results that help to easily classify definable sets (up to definable isomorphism) thanks to Euler characteristic. We will also introduce definable homotopies and prove some extension theorems. Our only reference will be [vdD98][Ch 8 §1, §2, §3 and Ch 6 lemmas 3.5 and 3.8].

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1. Some recalls

We begin this work by recalling the main tools we will use that have been proven in previous seminars to fix the the key definitions and to reference correctly each statement.

Notation 1.1. Let $(\mathcal{R}, +, *, 0, 1, <)$ be an ordered real closed field. Let $(\mathcal{R}, <, \mathcal{S})$ be the o-minimal expansion of such field. Throughout this work we will only consider this o-minimal structure to ensure to be in the hypothesis of the triangulation theorem.

Notation 1.2. For a given set $S \subseteq \mathcal{R}^n$ we denote it's topological closure with $\mathrm{cl}(S)$.

Definition 1.3. Let $\{a_0, ..., a_k\}$ a collection of k+1 points in $\mathcal{R}^m$. The k-*simplex spanned by* $\{a_0, ..., a_k\}$ is the set $(a_0, ..., a_k) := \{\Sigma_{i=0}^k t_i a_i |\ t_i \in \mathcal{R}_{>0}$ and $\Sigma_{i=0}^k t_i = 1\}$. We say a set $\sigma \subseteq \mathcal{R}^n$ is a k-*simplex* (or simply a simplex) if there exists a set of points $\{a_0, ..., a_k\}$ in $\mathcal{R}^n$ such that $\sigma = (a_0, ..., a_k)$. We say that $\{a_0, ..., a_k\}$ are the vertices of σ.

Remark 1.4. Simplexes are definable sets since they are semi-algebraic.

Remark 1.5. In this definition n-simplexes in $\mathcal{R}^n$ are open (differently from the classical definition).

Definition 1.6. Let $\sigma = (a_0, ..., a_k)$ be a simplex. We define the *barycenter* of σ as $\mathrm{b}(\sigma) := \frac{1}{k+1}\Sigma_{i=0}^k a_i$.

Definition 1.7. Let $\sigma = (a_0, ..., a_k)$ be a simplex. Let $i \in \{0, ..., k\}$. The *ith face* of σ is the simplex spanned by $\{a_0, ..., a_{i-1}, a_{i+1}, ..., a_k\}$. More generally, a *face* of σ is a simplex spanned by a subset of $\{a_0, ..., a_k\}$. If τ is a proper face of σ we write $\tau < \sigma$.

Definition 1.8. A *complex* in $\mathcal{R}^n$ is a finite collection, $K = \{\sigma_i\}_{i=0}^k$ of simplexes in $\mathcal{R}^n$ such that for any two $\sigma_i, \sigma_j \in K$, $\mathrm{cl}(\sigma_i) \cap \mathrm{cl}(\sigma_j)$ is either empty or a common face.

Definition 1.9. Let K be a complex. We define the *realization* of K to be the set $|K| = \cup_{\sigma \in K} \sigma$. We define $\mathrm{Vert}(K)$ to be the set of all vertices of simplexes in K.

Definition 1.10. Let K be a complex. A subcomplex of K is just a subset of K.

Remark 1.11. Let K be a complex. $|K|$ is definable since it's finite union of definable sets.

Remark 1.12. The datum of a complex K is equivalent to the datum of $\mathrm{Vert}(K)$ and which vertices span a simplex in K.

Definition 1.13. Let K be a complex. We denote by $\overline{K}$ the complex containing all the faces of the simplexes in K. We say K is a closed complex if it contains all the faces of its complexes (ie: $K = \overline{K}$).

Definition 1.14. Let K be a complex. Let $L \subseteq K$ be a subcomplex. We say L is a closed subcomplex of K if $\overline{L} \cap K = L$.

Remark 1.15. Let K be a complex. Let $L \subseteq K$ a subcomplex. It's trivial to check that $\mathrm{cl}(|K|) = |\overline{K}|$. We deduce that a L is a closed subcomplex of K if and only if $|L|$ is a closed subset of $|K|$.

Definition 1.16. Let K and L be two complexes in $\mathcal{R}^n$. A *vertex map* $V : K \to L$ is a function $V : \mathrm{Vert}(K) \to \mathrm{Vert}(L)$ such that for any collection of vertices $\{a_0, ..., a_k\}$ in $\mathrm{Vert}(K)$ that span a simplex in K, $(V(a_0), ...V(a_k)) \in L$.

Remark 1.17. Let K, L be two complexes. Let $V : K \to L$ be a vertex map between them. We can define:

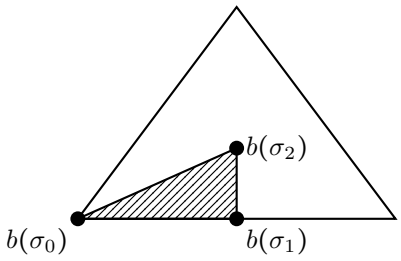
$$|V| : |K| \to |L|$$
$$\Sigma t_i a_i \mapsto \Sigma t_i V(a_i)$$

This function is semilinear and hence definable.

Definition 1.18. Let K be a complex in $\mathcal{R}^n$.

(1) A K-flag $\mathscr{F}$ is a sequence $\sigma_0 < ... < \sigma_f$ of simplexes in $\mathcal{R}^n$ such that $\sigma_f \in$K.
(2) $\mathrm{b}(\mathscr{F}) := (\mathrm{b}(\sigma_0), ...\mathrm{b}(\sigma_f)))$.
(3) The first barycentric decomposition of K is $K' := \{\mathrm{b}(\mathscr{F}) | \mathscr{F}$ is a K-flag$\}$.

Example 1.19. The following picture shows how the simplex spanned by a K-flag (bottom left point<lower face<whole triangle) defines a simplex of the barycentric decomposition:



Remark 1.20. Let K be a complex in $\mathcal{R}^n$. $|K| = |K'|$.

Definition 1.21. Let $A \subseteq \mathcal{R}^n$ be a definable set. A *triangulation* of A is a pair (ϕ, K) where, K is a complex in $\mathcal{R}^n$ and $\phi : A \to |K|$ is a definable homeomorphism. Let $A' \subseteq A$ a definable subset. We say that a triangulation (ϕ, K) for A is *compatible* with A' if A' is a disjoint union of elements of the set $\phi^{-1}(K) := \{\phi^{-1}(\sigma) | \sigma \in K\}$.

Theorem 1.22 ([vdD98] Ch 8 §2 Theorem 2.9)**.**] Let $S \subseteq \mathcal{R}^n$ be a definable set. Let $S_1, ..., S_k \subseteq S$ be definable subsets of S. Then S has a triangulation in $\mathcal{R}^n$ compatible with these subsets.

We end this section with two lemmas that will be useful in the last section of this work and revolve around partitions of unity.

Lemma 1.23 ([vdD98]Ch 6 Lemma 3.6)**.** Let $B \subseteq \mathcal{R}^n$ be a definable set. Let $A_0, A_1 \subseteq B$ be two disjoint definable closed subsets of B. Then there exist two definable open subsets $U_0, U_1 \subseteq B$ such that $A_i \subseteq U_i$, $i = 0, 1$ and $U_0 \cap U_1 = \emptyset$.

Lemma 1.24 ([vdD98] Ch 6 Lemma 3.8)**.** Let $(\mathcal{R}, <, \mathcal{S})$ be an o-minimal structure that expands a real closed ordered field. Let $B \subseteq \mathcal{R}^n$ be a definable set. Let $A_0, A_1 \subseteq B$ be two disjoint definable closed subsets of B. Then there exists a function $f : B \to [0, 1]$ definable and continuous such that $f^{-1}(i) = A_i$ for $i = 0, 1$.

2. Definable equivalence and homeomorphism types

In this section we will discuss some criteria to classify definable sets: up to definable equivalence and up to definable homeomorphism.

Definition 2.1. Let $A \subseteq \mathcal{R}^m$, $B \subseteq \mathcal{R}^n$ be two definable sets. Let $f : A \to B$ a definable function. We say that:

(1) f is a definable equivalence if there exists a definable function $g : B \to A$ such that $f \circ g = id_B$ and $g \circ f = id_A$.
(2) We say A and B are definably equivalent if there exists a definable equivalence $f : A \to B$.

Remark 2.2. In Definition 2.1 (1) is equivalent to f being bijective. If we denote the graph of f with Γ_f, by definition of definable map f is definable if and only if Γ_f is definable as set. We denote with f^{-1} the set theoretic inverse of f. $\Gamma_{f^{-1}}$ is obtained from Γ_f by switching places of its components and, therefore, is definable.

Theorem 2.3. Let A, B be two definable sets. They are definably equivalent if and only if $\mathrm{E}(A) = \mathrm{E}(B)$ and $\dim(A) = \dim(B)$.

Before proving this theorem we will discuss three facts that we will use in the next proof.

Lemma 2.4. Let σ be a k-simplex, $k \geq 1$.

(1) σ is disjoint union of 2 k-simplexes and a (k-1)-simplex.
(2) Any two k-simplexes are semilinearly equivalent
(3) $\mathrm{E}(\sigma) = (-1)^k$.

Proof. For (1) we use induction on k. For k=1: σ is a segment, we can consider its barycentric decomposition as in definition Definition 1.18. For the inductive step: if $\sigma = (a_0, ..., a_k)$, for any $i = 0, ..., k$ consider the ith face σ_i. By inductive hypothesis σ_i decomposes in $\tau_1 \sqcup \tau_2 \sqcup \delta$ where τ_i is a $(k-1)$-simplex (for $i = 0, 1$) and δ is a $(k-2)$-simplex. We obtain what we wanted if we consider the simplexes spanned by their respective vertices and a_i. (2) is trivial. For (3) we just need to use the fact that Euler's characteristic does not depend on the cell decomposition of a set and that simplexes are cells by induction since the standard simplexes are inductively related as follows:

$$\Delta_m := \{(x_0, ..., x_m, r) | (x_0, ..., x_m) \in \Delta_{m-1}, 0 < r < 1 - \Sigma x_i\}$$

We deduce that σ can be decomposed as just one k-cell. □

Proof. If A and B are definably equivalent we already know that they have the same dimension and Euler characteristic (it was proven when we discussed their well definition). For the other implication,

thanks to Theorem 1.22 both A and B are disjoint unions of simplexes.
We claim that A and B are semilinearly equivalent (hence definably equivalent). By induction on $k = \dim(A) = \dim(B)$. If $k = 0$ both A and B are a disjoint union of points and Euler's characteristic forces them to have the same cardinality. For $k > 0$: $A = \coprod_0^p \sigma_i \sqcup \coprod_{p+1}^q \sigma_j$ and $B = \coprod_0^r \tau_i \sqcup \coprod_{r+1}^s \tau_j$ where the σ_i and τ_i are the k-simplexes and the remaining parts have lower dimension. By Lemma 2.4(1) we can decompose σ_0 until the number of k-simplexes in the decomposition of A and B is the same (one k-simplex decomposes in two, increasing the number of the k-simplexes in the decomposition by one). Without loss of generality $p = r$. We can apply Lemma 2.4(1) once more to both sets so that in both decompositions there is at least one $k-1$-simplex. Now, by Lemma 2.4(2), there is a semilinear equivalence $\coprod_{i=0}^p \sigma_i \to \coprod_{i=0}^p \tau_i$ since all those simplexes are disjoint (so we just reduce to build a semilinear equivalence $\sigma_i \to \tau_i$ for all i). $A' := \coprod_{p+1}^q \sigma_j$ and $B' := \coprod_{p+1}^s \sigma_j$ are semilinearly equivalent by inductive hypothesis (we subtracted the same number of k-cells by Lemma 2.4(3)). We can assemble these two semilinear equivalences to one $A \to B$. □

We end this section by discussing definable homeomorphism types.

Theorem 2.5. Let $e_i \in \mathcal{R}^n$ denote the vectors $e_{i,j} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ otherwise} \end{cases}$. Let A be a definable set. A is definably homeomorphic to an union of faces of the standard simplex $(e_1, ...e_N) \subseteq \mathcal{R}^n$ for some N.

Proof. By Theorem 1.22 we know A is definably homeomorphic to a $|K|$, for some K complex. Let N be the cardinality of $\mathrm{Vert}(K)$. We can consider $V : \mathrm{Vert}(K) \to \{e_1, ..., e_N\}$ any bijective function. We can define a complex L by fixing $\mathrm{Vert}(L) := \{e_1, ..., e_N\}$ and by imposing that the simplexes of L are exactly the image of the simplexes of K. This construction provides a vertex map $V : K \to L$ which carries a continuous function $|V| : |K| \to |L|$. Notice that V^{-1} is again a vertex map by construction, therefore there is a continuous map $|V^{-1}| : |L| \to |K|$ that is clearly the inverse of $|V|$. □

Corollary 2.6. The set of definable homeomorphism types has numerable cardinality.

Proof. By Theorem 2.5 for any definable homeomorphism type there exists a representative collection of faces of a standard simplex. Standard simplexes are of numerable quantity and have at most a finite number of faces. We conclude that definable homeomorphism types are at most a numerable union of finite sets ergo numerable. All standard simplexes are not definable homeomorphic to each other (they have different dimension) so definable homeomorphism types are at least numerable. We conclude by Schröder–Bernstein theorem. □

Discussion 2.7. We point out that the proof of Theorem 2.3 does not construct a definable homeomorphism since we can assemble maps from disjoint union to a definable function of sets but topology does not behave well when we assemble the map on k-simplexes with the map on the lower dimension simplexes. The broadest example could be:

$$(0, 1) \sqcup \{300\} \to (0, 1]$$

given by the identity on $(0, 1)$ and $300 \mapsto 1$.

3. Definable homotopies

In the following section, we give the definition of definable homotopy. Most of the section will focus around Proposition 3.5 and we will use it to prove Corollary 3.16 which lets us extend definable continuous maps from a closed subset to a definably contractible space.

Definition 3.1. Let K be a complex. Let A be a definable subset of $|K|$. We define the (K-)*star* of A as the following set: $\mathrm{st}_K(A) := \bigcup_{\mathrm{cl}(\sigma)\cap A\neq\emptyset,\sigma\in K}\sigma$

Remark 3.2. The $K-$star of a set is always an open definable (is finite union of definable sets) subset of K.

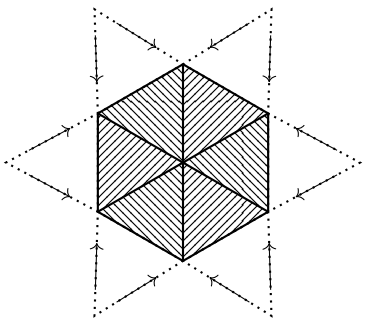
Definition 3.3. Let S,T be two definable sets. Let $f,g : S \to T$ be two definable continuous maps. A definable homotopy between f and g is a definable continuous map $H : S \times [0,1] \to T$ such that $H(s,0) = f(s)$ and $H(s,1) = g(s)$. We write $f \sim_h g$.

Definition 3.4. Let S,T be two definable sets. Let $r : S \to T$ a definable continuous map.

(1) r is a definable retraction if there exists a definable contiuous map $s : T \to S$ such that $r \circ s = id_S$.
(2) r is a definable deformation retraction if it is a retraction and there exists a definable homotopy between $s \circ r$ and id_R.
(3) r is a definable strong deformation retraction if it is a deformation retraction and, moreover, one of the homotopies in the definition fixes S.

Proposition 3.5. Let K be a complex. Let $L \subseteq K$ a closed subcomplex. Then there exists a definable retraction $r : \mathrm{st}_{K'}(|L|) \to |L|$ such that for all x in $\mathrm{st}_{K'}(|L|) - |L|$ the segment $(x, r(x)) \subseteq \sigma \in K'$

Example 3.6. The following picture exemplifies the retraction in the proposition around $\mathrm{st}_{K'}(|L|)\|L|$.



Remark 3.7. In the hypothesis of Proposition 3.5 we can define a definable homotopy $H(X,t) = tx + (1-t)r(x)$ which exhibits r as a definable strong deformation retraction.

We begin to build the machinery to prove Proposition 3.5.

Construction 3.8. We assume to be in the hypothesis of Proposition 3.5.

(1) $E := \mathrm{Vert}(K) = \mathrm{Vert}(K)$.
(2) Let $e \in E$ be a vertex. For any $x \in |\overline{K}|$ then $x \in \sigma = (e_0, ..., e_k)$ (since K is a complex sigma is unique) for some $\sigma \in K$ and $x = \Sigma t_i e_i$.
We denote with $\lambda_e : |\overline{K}| \to [0,1]$ the function:
$$x \mapsto \begin{cases} 0 & \text{if } e \notin \{e_0, ...e_k\} \\ t_i & \text{if } e = e_i \text{ for some } i \end{cases}$$
(3) Let σ be an element of K. $\lambda_\sigma := \lambda_{\mathrm{b}(\sigma)} : |\overline{K}| = |\overline{K'}| \to [0,1]$
(4) We denote with $w_\sigma : |\overline{K}| \to [0,1]$ the function:
$$x \mapsto \begin{cases} 1 & \text{if } \sigma \in L \\ 0 & \text{if } \sigma \in \overline{K} - \overline{L} \text{ or } (*) \\ \frac{\Sigma_{\sigma<\tau\in L}\lambda_\tau(x)}{\Sigma_{\sigma<\tau\in K}\lambda_\tau(x)} & \text{if } \sigma \in \overline{L} - L \end{cases}$$

$(*) : x \in \overline{L} - L$ and $d_\sigma(x) := \Sigma_{\sigma<\tau\in K}\lambda_\tau(x) = 0$

(5) $\mu_s := \lambda_\sigma * w_\sigma := |\overline{K}| \to [0,1]$

Remark 3.9. All the maps above are definable. Notice that $\Sigma_{e\in E}\lambda_e(x) = 1$ since the non empty terms coincide with x's affine coordinates. w_σ might be discontinuous due to $(*)$. Lemma 3.10 shows that μ_σ is continuous.

Lemma 3.10. Let the hypothesis of Proposition 3.5 be verified. Let σ be a simplex of $\overline{L} - L$. Let S be an element of K' such that for all $s \in S$ $d_\sigma(s) = 0$ then $\lambda_\sigma(s) = 0$ for all $s \in S$.

Proof. We know $S = (\mathrm{b}(\sigma_0), ..., \mathrm{b}(\sigma_n))$ for $\sigma_0 < ... < \sigma_n \in K$ a K-flag. By contradiction, let us assume that there exists an $s \in S$ such that $\lambda_\sigma(s) \neq 0$. We deduce, by definition of λ_σ that $\mathrm{b}(\sigma) = \mathrm{b}(\sigma_i)$ for some i, hence $\sigma = \sigma_i$ for some i (K is a complex). Let us assume that $\sigma = \sigma_n$. $\sigma_n \in K$ by definition of K-flag but $\sigma \in \overline{L}$ therefore $\sigma \in \overline{L} \cap K = L$ since L is closed. Since we assumed $\sigma \in \overline{L} - L$ we arrive at a contradiction. We deduce that $\sigma = \sigma_i < \sigma_n$ for $i < n$. If that is the case then $\tau := \sigma_n \in K$ and $\lambda_\tau(s) \neq 0$ (we know that $\mathrm{b}(\tau) = \mathrm{b}(\sigma_n)$ is a vertex of S). Since all terms of $d_\sigma(s)$ are nonnegative, we deduce $d(s) \neq 0$. □

Lemma 3.11. Let us assume to be in the hypothesis of Proposition 3.5. Let x be any point in $\mathrm{st}_{K'}(|L|)$. There exists a $\sigma \in L$ such that $\mu_\sigma(x) > 0$.

Proof. We show that for any simplex $S \in K'$ such that $\mathrm{cl}(S) \cap |L| \neq \varnothing$ there exists a σ such that $\mu_\sigma(s) > 0$ for all $s \in S$. As before $S = (\mathrm{b}(\sigma_0), ..., \mathrm{b}(\sigma_n))$ for $\sigma_0 < ... < \sigma_n \in K$ a K-flag. By definition of S, it intersects non trivially a simplex of L. Since S is a simplex of K' and L is a subcomplex of K, $cl(S) \cap L$ contains a face of S. We deduce that one of the vertices of S is contained in a simplex of L and therefore it is its barycenter. We deduce that one of the σ_i lies in L. Let's fix $\sigma = \sigma_i$ for such i. We know, by definition of w_σ, that it is constant $w_\sigma = 1$. We conclude that $\mu_\sigma(s) = w_\sigma(s)\lambda_\sigma(s) = \lambda_\sigma(s) > 0$ since for all $s \in S$ since $\mathrm{b}(\sigma)$ is a vertex of S. □

Construction 3.12. We can now give a constructive definition for the retraction in Proposition 3.5:

$$r : \mathrm{st}_{K'}(|L|) \to \mathcal{R}^n$$
$$x \mapsto \frac{\Sigma_{\sigma\in\overline{K}}\mu_\sigma(x)\mathrm{b}(\sigma)}{\Sigma_{\sigma\in\overline{K}}\mu_\sigma(x)}$$

It's clearly definable and continuous due to Lemma 3.10.

Next lemma actually proves *Proposition* 3.5.

Lemma 3.13. Let us assume to be in the hypothesis of *Proposition* 3.5. Let r be map in Construction 3.12. The followings hold:

(1) Let $S \in K'$ be such that $S \subseteq \mathrm{st}_{K'}(|L|)$. Then $r(S) \subseteq \mathrm{cl}(S) \cap |L|$.
(2) Let x be a point of $|L|$. Then $r(x) = x$.

Proof. As before, we know $S = (\mathrm{b}(\sigma_0), ..., \mathrm{b}(\sigma_n))$ for $\sigma_0 < ... < \sigma_n \in K$ a K-flag. As for Lemma 3 we know there exists an $i \in \{0, ..., n\}$ such that $\sigma_i \in L$. Let k be the maximum of such. Let us consider now any $\sigma \in \overline{K}$ such that $\mu_\sigma \neq 0$ on S (being non-zero on one point implies being non-zero over all S by the same steps of Lemma 3). $\mu_\sigma \neq 0$ on S implies $\lambda_\sigma \neq 0$ on S which implies that $\sigma = \sigma_t$ for some t. If $t > k$, for any $\tau > \sigma > \sigma_k$ with $\tau \in L$ and $\lambda_\tau \neq 0$ on S we would have $\sigma_k < \tau$, which would contradict the maximality of S. We deduce then that the numerator of $w_\sigma(s) = \Sigma_{\sigma<\tau\in L}\lambda_\tau(x) = 0$ for all $s \in S$ which contradicts our assumption on σ. We deduce that we can rewrite Construction 3.12 as:

$$r(x) = \frac{\Sigma_{0\le i\le k}\mu_{\sigma_i}(x)\mathrm{b}(\sigma_i)}{\Sigma_{0\le i\le k}\mu_{\sigma_i}(x)} = \Sigma_{0\le i\le k}\alpha_i b(\sigma_i)$$

for some $\alpha_i \neq 0$ with $\Sigma\alpha_i = 1$. We observe from that that $r(x)$ is a point in the simplex spanned by $\{\sigma_0, ..., \sigma_k\}$ which is contained in $\mathrm{cl}(S)$ and in $\sigma_k \subseteq |L|$. We have therefore proven that $r(S) \subseteq \mathrm{cl}(S) \cap |L|$. Now we compute r when $S \subseteq |L|$. Then $\sigma_n \in L$ and $r = n$. We claim that for all σ $\mu_\sigma = \lambda_\sigma$ on S.

(1) If σ in L we are already done by definition of w_σ.
(2) If $\sigma \in \overline{L} - L$ we have to compute w_σ. Let τ be any simplex in K such that $\lambda_\tau \neq 0$ on S and $\tau > \sigma$. Since $\lambda_\tau \neq 0$ on S we deduce that $\mathrm{b}(\tau)$ is a vertex of $S \subseteq |L|$ and, therefore, $\tau \in \overline{L}$. $\tau \in \overline{L} \cap K = L$, since L is a closed subcomplex. This shows that the numerator of w_σ is equal to the denominator since they have the same non 0 terms.
(3) If σ in $\overline{K} - \overline{L}$ we have $w_\sigma = 0$. By an analog reasoning to (2) we realize that $\lambda_\sigma = 0$ on all S.

Let now x be an element of $|L|$. The claim lets us rewrite $r(x)$ as:

$$r(x) = \Sigma_{i=0}^{n} \lambda_{\sigma_i}(x)\mathrm{b}(\sigma_i) = x$$

by Remark 3.9. □

Corollary 3.14. Let $B \subseteq \mathcal{R}^n$ be a definable set. Let $A \subseteq B$ a closed definable subset. Then there exists an open definable set $U \subseteq B$ such that $A \subseteq U$ and there exists a definable retraction $r : \mathrm{cl}(U) \cap B \to A$.

Proof. We begin to apply Theorem 1.22 to $A \subseteq B$ which provides us with a triangulation, (ϕ, K), for $\mathcal{R}^n$ relative to B and A. Applying ϕ produces two complexes $L \subseteq K$ such that $|L| \cong A$ and $|K| \cong B$ with L closed in K. By Proposition 3.5 we know there is a definable retraction $\mathrm{st}_{K'}(|L|) \to |L|$. Through ϕ we recover that there exists an open definable set $A \subseteq V \subseteq B$ such that $r : V \to A$. Now We apply Lemma 1.23 to A and $B - V$ which provides two definable open disjoint subsets $A \subseteq U_0 \subseteq B$ and $B - V \subseteq U_1 \subseteq B$. Now $U_0 \subseteq \mathrm{cl}(U_0) \subseteq \mathcal{R}^n - U_1 \subseteq \mathcal{R}^n - (B - V) = B \cap V$. We set $U := U_0$ and the desired retraction is $r_{|U}$. □

Definition 3.15. Let C be a definable set. Let x be an element of C. We say that C is definably contractible if $C \to \{x\} \hookrightarrow C$ is definably homotopic to the identity of C.

Corollary 3.16. Let B be a definable set. Let $A \subseteq B$ be a closed definable subset. Let C be a definably contractible space. Let $f : A \to C$ be a definable continuous map. Then there exists a definable continuous map $\overline{f} : B \to C$ such that $\overline{f}_{|A} = f$.

Proof. Let U and $r : \mathrm{cl}(U) \cap B \to A$ be as in Corollary 3.14. By Lemma 1.24 there exists a $\lambda : B \to [0, 1]$ such that $\lambda^{-1}(0) = A$ and $\lambda^{-1}(1) = B - U$. Let c be a point in C. Let $\phi : C \times [0, 1] \to C$ be a definable homotopy between id_C and $const_c$. We define $\overline{f} : B \to C$ as the following function:

$$\overline{f}(x) := \begin{cases} \phi(f \circ r(x), \lambda(x)) & x \in \mathrm{cl}(U) \cap B \\ c & otherwise \end{cases}$$

This function is clearly definable (it's graph is finite union of definable sets). It clearly restricts to f and is continuous. □

References

[vdD98] Lou van den Dries. *Tame topology and o-minimal structures*. Number 248 in London Mathematical Society Lecture Note Series. Cambridge University Press, 1998. MR:1633348. Zbl:0953.03045. 1, 2, 3